

A New Wave Hits California Swimming Pools

*Operating pool pumps for longer periods at slower flow rates
can adequately filter the water while cutting the energy use in half.*

by Scott Clay

A commonly held view in the pool and spa industry is that since September 11, 2001, homeowners have reduced their spending on vacation travel and are putting the hard-earned dollars that they would have spent on airfare and motels into their homes instead. A large portion of this spending has gone to improving their backyards. "The backyard is now becoming the safe haven for those who previously traveled to exotic places. The swimming pool is usually the center focus of the 'total backyard,'" says Phil Gelhaus of General Pool Supply, in Rancho Cordova, California. The total backyard typically consists of a swimming pool, spa, patio, built-in barbecue center, entertainment system, and sometimes a pond, fountain, and waterfall.

While the total backyard may offer an easily accessible pleasant retreat, it is expensive to operate and maintain. Swimming pools, spas, fountains, ponds, and waterfalls all require the use of electric pumps to keep the water flowing. With current electric rates reaching almost \$0.24/kWh for large residential energy consumers in California, the typical backyard paradise can cost \$150–\$200 per month, not counting lighting and natural gas for the barbecue. Of course the largest energy hog in your backyard getaway is usually the swimming pool.



Scott Clay (left) and Gary Fernstrom of PG&E prepare a pool pump demonstration for pool service professionals.

Pumping Up the Energy Savings

California swimming pools average 25,000 gallons, and they are typically fitted with a 1 1/2 hp filtration pump. Most new pools are built with a unit that is capable of pumping the entire volume of the pool through the filtration system in eight hours. Pool builders instruct new pool owners to run their filtration pump six to eight hours each day to maintain a clean pool. But the cost of energy to operate a 1 1/2 horsepower pump eight hours a day can reach \$110 per month. It's no wonder that California pool owners are searching for ways to reduce their pool-related energy costs.

To address this problem, pool service professionals have been installing smaller pumps when the original pump eventually fails. Since small pumps use less energy to operate than large pumps,

replacing the original filtration pump with a smaller one should save both energy and money. Many pool service professionals in California, in order to save energy, have been replacing failed pool pumps with pumps that are up to 50% smaller. But smaller pumps move less water and therefore must be operated longer to filter the same amount of water as a large pump. And not increasing pumping time to account for the reduction in water filtration can lead to unhealthy pools. So if pool owners want a clean pool, how much energy can they actually save by switching to a smaller pump?

The answer to this dilemma is to pump the water more slowly. "Our motto when talk-

ing about pool filtration is 'slower and longer is better,'" says Gary Fernstrom, senior program manager, who provides technical support for the Pacific Gas and Electric Company (PG&E) Pool Pump and Motor Rebate program. According to the pump affinity law, or pool pump cube rule, when the speed of the pump motor is lowered to one-half of the original rpm, the power demand of that pump motor scales with the cube of the speed of the motor. In other words, theoretically, when you drop the speed of a pump motor by half, you use one-eighth the power. This same rule of physics also states that the flow of water from a pump whose motor speed has been reduced by half is proportional to the speed of the motor—half the motor speed will result in half the water flow.

Slowing the pump motor to half of the original speed will require the opera-

tor to operate the pump twice as long to pump the same amount of water, but the motor is drawing only one-eighth the power. So one-eighth the power multiplied by twice the time, theoretically, yields one-quarter of the energy use for the same water turnover. In practice, this turns out to be about one-third of the energy use, given the performance of real-world pumps and motors (see "Other Ways to Save").

Are there pool pumps that operate at half speed? Yes, there are! They are called two-speed or low-speed pool pumps. All major manufacturers of pool pumping equipment make two-speed pumps. When a two-speed pump is operating on high speed, the motor turns at 3,450 rpm, the same speed as the typical residential single-speed pool pump. When the pump is operating on low speed, the motor rotates at 1,725 rpm, exactly one-half of the high speed. The high speed of the two-speed pump is used in situations where a higher flow is required, such as when a pool owner is using a pressure side pool sweep or lifting pool water up to solar heating collectors on the roof of the house. But in general, the low speed is used for most of the filtration cycle. Moving the water more slowly and for a longer period of time uses significantly less energy and dramatically lowers operating costs—often by more than 50% for the typical swimming pool in California.

So why are two-speed pumps not being routinely used on swimming pools? Unfortunately, the common

belief is still that larger pumps pump faster, ending the pumping cycle sooner and therefore using less energy. Convincing the pool industry that slower and longer is better is a matter of education.

So in October 2002, PG&E built a mobile pool pump demonstration unit to show the pool industry that low-speed pumping uses a lot less energy. The demonstration unit provides irrefutable evidence that using large pool pumps and pumping for shorter periods is significantly more costly than using small

thousands of customers and pool professionals at home and garden shows, pool industry trade shows, and pool trade association meetings over the last two years.

The two-speed pool pump is becoming more common on new pools. Terry LaRocca of Perfection Pools in Chico, California, says, "I install only two-speed or low-speed pool pumps on all of my new pools. Two-speeds bring the cost of filtering your pool back into the real world. The pool owners love the energy savings." Robert Livingston of Emerald Pools in Fresno, California, adds, "All my new pools are designed with two-speed or low-speed pumps. The owner can filter the water for up to 12 hours a day on low speed and still save energy and electrical costs." (Many pool contractors feel, as this pool builder does, that it's not necessary to filter the entire volume of pool water every day to maintain a clean pool.) Jeff Smart of Holiday Pools, in Chico, California, who builds pools throughout California, says with pride, "Holiday Pools has been saving our customers money by using two-speed pool pump technology on our new pools for years."



Above each of the variously-sized pool pumps on the demonstration trailer are controls. The box on the upper right displays flow rates and energy use for individual pumps.

pumps and pumping longer, and that the energy consumed when using the low speed of a two-speed pool pump is only a small fraction of that used when pumping on high speed. When they witness this first hand, contractors and customers alike are convinced. The mobile unit has demonstrated these energy savings to

Aggregate the Savings

These pool builders share a common goal. That goal is to build new pools that owners can afford to operate. In addition to saving their customers money, they are also helping to significantly reduce California's electrical energy demand. There are approximately 1.2 million residential, privately owned in-ground swimming

Other Ways to Save

In addition to reducing the pump horsepower, pool owners can save energy by keeping the filters clean and using a pool cover when the pool is not in use. These are excellent ways to further reduce operation costs, while keeping the pool clean and swimmers healthy.

Clean filters reduce the system pressure against which the filtration pump must pump. Residential pool filter

systems are equipped with pressure gauges that measure the differential pressure between the inlet and outlet of the filter system. Each manufacturer provides differential pressure specifications so that pool owners or service professionals can determine when to clean the filters. Strict adherence to these specifications will save energy, since a clogged filter will decrease pump flow and require the pool

owner to run the pump longer to maintain a clean pool.

Pool covers prevent dirt and debris that would need to be filtered from entering the pool, thereby reducing daily pumping time. Also, according to the DOE's Office of Energy Efficiency and Renewable Energy, pool covers reduce evaporation, which cools pool water, and can save up to 70% of the cost of heating a pool.

Appliances

pools in California. (Above ground pools are not included in this discussion. Typically, above-ground pools are not filtered as routinely as in-ground pools.) Reducing the power demand by two-thirds and the energy use from pool pumping by half can help the state to better manage its power grid. This is especially important at critical times, such as energy shortages like the one in the summer of 2001.

What is the state of California doing to support energy reduction by pool owners? Through utilities such as PG&E, the state provides \$300 rebates to owners who replace their pool filtration pumps with two-speed or low-speed pumps. With a statewide program that encourages slow pool filtration, the new wave in pool pumping is sure to change the way we maintain our pools. Finally, the cost of using that backyard paradise does not have to equal the cost of an exotic vacation.



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For more information:

The DOE's Office of Energy Efficiency and Renewable Energy has helpful information about pools and energy efficiency at www.eere.energy.gov/rspec.

The California Spa and Pool Education Council has Pool Owner Tips at www.calspec.org.